Bloody robots as emotional design. How emotional structures may change expectations of technology use in hospitals

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Abstract

By applying Gilles Fauconnier & Mark Turner's theory of *conceptual blending* to a design case I demonstrate how experiencing emotional qualities in technology design may influence the way users cognitively reconstruct standard expectations of use. In so doing, I expand the dominating cognitive theory of emotion in design in three central respects: (i) the understanding of mixed emotions is deepened; (ii) a more detailed explanation is given of the specific operations involved in appraisal processes grounded in embodied interaction; (iii) a structural model is proposed for mapping the constitutive role that mixed emotions play in product usage and interaction.

Conference Themes: Modeling Experience, Usage and Interaction, Theoretical Issues

Keywords: User Experience, Emotional Robots, Cognitive Semiotics, Appraisal Theories of Emotion

Introduction

Emotional approaches to technology design have proved very promising for the ongoing proliferation of digital technology into 'soft' sectors within healthcare centers and hospitals. As information technologies are gradually getting responsible for more direct patient-related tasks (as in *pervasive healthcare*), one overarching goal of HCI-designers working in these domains has been to find methods of incorporating considerations for patient's emotions into the design of interactive healthcare systems. This practice could be described as part of a broader *emotion-driven* design paradigm to use a term from Desmet & Dijkhuis (2003).

Emotion-driven design in healthcare sectors differs from other emotional approaches in that the designer often needs to have a more nuanced understanding of *mixed emotions*. Medical treatment typically implies that patients live through compound or even conflicting emotional states. The treatment itself might involve unpleasant affective states, yet at the same time patients tend to appraise these stimuli as the

'necessary evil' for achieving a beneficiary goal: cure and well-being. Such instances show that user experience is not perceived to be 'valenced' in a clear-cut *bipolar* way (either good or bad, pleasant or unpleasant), which some models of emotional principles could give the impression of (cf. Russell, 1980).

Moreover, patients also have to adapt emotionally to the new functions and circumstances of technology use for healthcare purposes. Basically, this adaptational process requires that standard expectations of use are revised according to the emotional reactions caused by novel or changing eliciting conditions. To facilitate the implementing of new emotionally engaging technologies, designers could then benefit from gaining further insight into how our bodily feelings, emotions and cognitive operations mutually shape one another.

Recently, we have witnessed a boom in the development of emotion theories in design (e.g. Desmet, 2002; McDonagh, Hekkert, van Erp & Gyi, 2004). However, to my knowledge, mixed emotions and, in particular, the eliciting of mixed emotions by the appraisal of novel experiential and embodied aspects of technology use has not received the attention it deserves. For instance, Desmet's (2002: 124) 'basic model of product emotions' is developed primarily with regard to 'passive observation', not embodied interaction. Also, in the research literature novelty appraisals are generally reduced to being merely a question of products and interfaces evoking surprise, fun or pleasure (e.g. Green & Jordan, 2002; Desmet & Dijkhuis, 2003), whereas the emotionally motivated reshaping of standard knowledge structures in the user is left largely unspecified.

The aim of this paper is therefore to increase designer's understanding of the eliciting of mixed emotions by the appraisal of novel embodied aspects of interactive healthcare design. For that purpose I will apply Gilles Fauconnier & Mark Turner's theory of conceptual blending (Fauconnier & Turner, 1998, 2002) to a design case.

Blending theory posses an unexplored potential to fill in those gaps in the existing appraisal theoretical framework that Roseman & Smith (2001) have pointed out. First, it offers a more elaborated account of human's ability to construct mental representation (appraisals, interpretations, judgments), which appraisal theories generally hold to be pivotal for our emotional and bodily transactions with the world (cf. Lazarus, 1991). Secondly, it provides structural models that enable us to articulate the specific operations by which new appraisals and evaluations might occur. These two obvious heuristic values will stand out more clearly as I move from Desmet's model of product emotions to Fauconnier & Turner's network model in my subsequent case analysis of Roblood.

1. ROBLOOD

RoBlood is a series of blood-taking robots that is meant to replace bio-analysts in the Danish healthcare sector. Since 1987 blood tests in this sector have increased at a rate of 7% per year amounting to the total of ten million tests in 2007. Because of this increase, bio-analysts are today using 40% of their working hours in actual taking blood. Besides raising the question of whether it is a reasonable use of high trained analysts for this rather simple task, the problem is also that the repetitive nature of blood-taking is causing a lot of arm, hand and finger injuries (Wetton, 2007).

Since blood-taking is a very personal and intimate affair, the ultimate goal of the RoBlood-project was to design robots that patients felt emotionally secure and safe about. But how do you make people feel comfortable when intimate human interaction is handed over to a robotic device, which many of us happens to associate with hostile cyborgs and androids chasing the existence of mankind in sci-fi films such as *Terminator* and *The Matrix*. Is it possible at all to engender the same sense of confidence in a robot around the emotional issue of people's well being?



Fig. 1. Sessio – a bloodtaking robot

The strategy of the design team was to push the 'soft' and 'human' aspects of the workspaces in hospitals to the fore. This strategy is clearly reflected in the way in which technology use, emotions and form are unified in one of the design proposals named *Sessio*. *Sessio* integrates blood-taking robotic technology into the armrest of an

organically shaped interactive chair (fig. 1). When the patient places her arm on the armrest, it automatically adjusts in height and follows the patient's movements. Then a comfortable vacuum surrounds the arm, so that it is enclosed and kept tranquil. The blood-taking starts, when after the position and depth of veins have been identified, a needle from the armrest is inserted into the patient's arm (cf. Nørhave, Madsen & Springborg, 2007).

2. Limitations of the Appraisal Approach to Emotion

Now, the interesting question to ask is: What kind of eliciting process of emotions is involved here? According to Desmet (2002: 111) "the key to understanding the eliciting conditions of distinct emotions lies in the characteristic of the appraisal process". An 'appraisal' is conceived as a mental judgment or evaluation of whether or not a particular product is beneficial, harmful or not relevant to our well-being. Appraisals are made on the basis on the perception of sensuous qualities of the product, but are also guided by internal phenomena such as our drives, needs, instincts, motives, goals, values and so fort. All of which is covered by the term *concern* (cf. Frijda, 1986).

In order to illustrate how these components intertwine in the eliciting process, Desmet presents his basic model of product emotions (see fig. 2). He further argues that product emotions can be modeled according four major appraisal types: Appealingness, Legitimacy, Motive Compliance and Novelty.

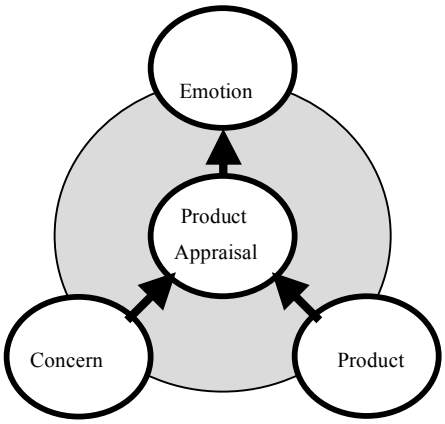


Fig. 2 Desmet's basic model of product emotions (Adapted from Desmet, 2002)

If we accept this model for the time being and follow the principal methodological guidelines of its use, as they are laid out by Desmet, we are able to distinguish between different product emotions involved in the *Sessio* experience. (For the sake of clarity in argument I leave appraisal of Legimitacy and Motive Compliance out of consideration, since they are not all that relevant for my purpose).

First, there is the appraisal of Appealingness (fig. 3a). If we assume that most people are positively disposed toward organic and round shapes, then it would be reasonable to claim that the mere visual appearance of *Sessio* is likely to evoke an appraisal of appealingness in the majority of patients. The key idea of appraisal theories is then that such an appraisal is able to make patients feel pleasant or good about the technology design.

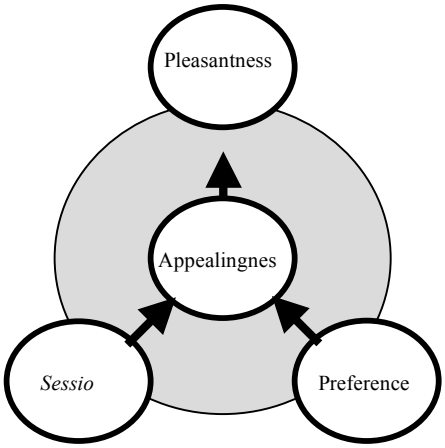


Fig. 3a Appraisal of Appealingness

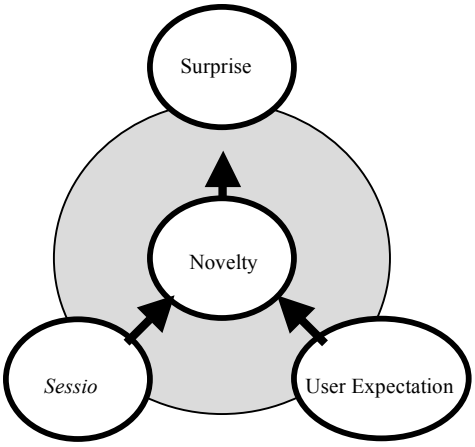


Fig. 3b Appraisal of Novelty

Secondly, there is the appraisal of Novelty (fig. 3b). According to Desmet (2002: 117) novelty appraisal is not primarily related to a particular concern type, but to "our knowledge and expectations". It arises as a result, when we meet unfamiliar products that deviate from what we already know or have learned to expect through previous experiences. Surely, *Sessio* is a good candidate for evoking this appraisal type as it is designed for functionalities that robots are not normally expected to perform. Robots are most often employed for manipulating inanimate objects at, for example, the assembling lines within the car industry, not for sensitive and careful operations on human skin. For that reason, the immediate emotional response to follow from user's cognitive evaluation of *Sessio* is surprise.

Obviously, these brief illustrations are by no means doing justice to Desmet's otherwise exhaustive framework. Yet it does provide sufficient material for discussing two central challenges that I believe appraisal theories of emotion in design are currently faced with.

The Complex Emotional Structure of Embodied Interaction

Desmet is himself very explicit about the first challenge. As his model is based on passive observation as the primary perceptual mode, it has serious difficulties explaining external eliciting conditions involving embodied interaction and actual use in addition to visual experience. We thereby get an incomplete picture of emotional user experience as *Sessio* provides us with a clear example of. Consider again the appraisal of appealingness. As soon as patients actually start interacting with *Sessio*, they will experience quite the opposite of pleasantness. Because, in the actual use, the healthcare design does not *feel* the way it *looks*. Thus, the experience of the rounded shapes and smooth embrace by vacuum is rather sharply contrasted with the feeling of pain caused by the sting of a needle. Pleasantness then suddenly turns into unpleasantness.

Cases like this reveals the importance of extending the model to include the perceptual detection of changes in bodily states as well as the co-activation of competing emotions following from embodied interaction with a product.

There is one particular reason why Desmet's model, in its current set-up, is unable to provide us with that solution. That is because it 'freezes' product emotions as isolated and fixed moments in a linear sequence of user experiences (cf. fig. 3a, 3b). Consequently, mixed emotions are thereby not treated as mixed, but as one emotion being

temporarily superseded by another emotion. However, to account for the emotional profile of healthcare design products like *Sessio*, we must be able to describe how conflicting emotions are co-activated and fused through the experiential process of use. (Admittedly, Desmet deals with mixed emotions in his development of the Product Emotion Measurement instrument PrEmo, but they are not sufficiently treated in his basic model of product emotions).

Embodied Appraisals

The second challenge to be dealt with here consists, as Damasio (1994) has argued, in describing the role that emotions play in the active shaping of mental states and representations. More specifically, we need to account for what this co-activation of competing emotions means for the way we make sense out of products. The direction of arrows used in Desmet's model clearly reflects the opposite and seemingly inconsistent idea in appraisal theories, viz. that emotions are first and foremost *preconditioned and caused* by mental representations, not vice versa (cf. Lazarus, 1991).

Mental representations are no doubt largely responsible for the eliciting of emotions in user experience. For instance, if patients in the hospital are told that a robot named *Sessio* will take care of their blood test, then many of them are likely to have a concept of a robot. Moreover, this concept will on many occasions trigger emotions such as anxiety or even fear. Naturally, culture has a huge impact on the content and structure of such concepts. The not exactly flattering portraying of robots in mainstream film and entertainment might have led to the forging of an unconscious link between anxiety and robots in people's long-term memory. However, it is crucial to notice that the emotions arising from bodily interaction with *Sessio* prompt users to transform these standard expectations according to a new emotional content (see below).

In order to describe how emotions may in this way influence the reorganization of existing mental knowledge structures, I believe with Prinz (2004) that there is a need to supply the concept of 'cognitive appraisals' with a concept like 'embodied appraisals'. Prinz suggest 'embodied appraisals' as a new concept for explaining how not only mental states, but also *perceptual states* involved in the detection of bodily changes might cause the eliciting of emotions. As we will see shortly, this is a critical step toward acknowledging the role that emotions play in learning and adaptational processes. Of course, a structural model of emotions in design must also account for this aspect.

To sum up, in the reminder of this paper I want to focus on these three requirements to an appraisal model of emotions in design:

- Encompassing embodied interaction and use in addition to visual experience
- Clarifying how emotional conflicts are co-activated
- Modeling how mixed emotions might change user expectations through embodied appraisals

3. The Conceptual Integration of Emotion and Cognition

In what follows I will argue that all these requirements are met by Fauconnier & Turner's blending theory (Fauconnier & Turner, 1998, 2002). So, even if it is traditionally counted as belonging to the branches of cognitive semiotics and cognitive linguistics, blending theory also contains significant insights for appraisal theories of emotion in design. Before going into a more detailed discussion of these issues, let me just briefly introduce some necessary building blocks.

Blending theory is essentially based on the notion of 'mental spaces'. A mental space can be seen as a 'conceptual package' constructed in our mind *as* we think and talk for purposes of local understanding and action (cf. Fauconnier & Turner, 2002: 40). A mental space is thus built up dynamically in *working memory* out of salient elements and relations given through our perceptual and bodily interactions with the physical world. But in its organization of this experiential content, a mental space equally draws upon schematic structures available from *long-term memory* such as frames and image schemas. More clarity can be cast on mental space construction, if we turn toward the context of our case example.

Upon entering a hospital for a blood test we normally experience waiting rooms, nurses chatting in corridors, hospital porters transporting patients between operation and bed sections, the distinct smell of liquids for disinfection and medicaments, and so forth. All sorts of experiential information cueing us to construct a mental space of our purpose for being there. Additionally, we may employ conceptual structures of expectation acquired through similar experiences in the past, which specify the nature of relevant activity, events, and roles for interpersonal transaction (cf. Fauconnier & Turner, 2002: 104). Presumably, most Danish patients have become familiar with blood-taking through Denmark's public vaccination program. Therefore, it is reasonable to believe that a large

group of patient's posses a fairly well developed set of expectations. This set might include elements such as *healthcare professional, syringe, patient, a body, blood samples*, etc. Similarly, these elements can be organized according to a conceptually 'scripted' (Schank & Abelson, 1977) sequence of blood-taking actions such as *identification of veins, insertion of syringe, analysis of blood sample, diagnosis, treatment*, and so on. When the elements and relations of a mental space are organized in this way "as a package we already know, we say that the mental space is *framed* and we call that organization a *frame*" (Fauconnier & Turner, 2002: 102).

All of this is actually well known. What is less known is how framed mental spaces get modified under the pressure from incoming information of local contexts such as unfamiliar sensuous and emotional aspects of new artifacts. In other words: What happens to a mental space when we see and use a product that does not quite fit into our conceptual structures of expectation? One of the major achievements of Fauconnier & Turner is to have discovered that such experiences are governed by the principles of what they call 'conceptual blending'. Let's see how this applies in the case of *Sessio*.

Modeling User Experience

As user experience unfolds, a rich array of mental spaces is typically set up with mutual connections between them (Fauconnier, 2007: 352). In my attempt to unravel the interplay of emotion and cognition in the *Sessio* experience, it is useful to distinguish between two basic mental spaces.

Perceiving the visual form of the device (cf. Desmet's 'product appearance') means extracting salient features that are provisionally stored in a mental space. From this users are able to recognize the shape of a chair, but probably not to infer that the sitting function of this chair is combined with a blood-taking function. However, provided that they are informed about its primary use, and the richness of contextual clues, it is more than likely that users will frame this product experience by a standard BLOOD-TAKING frame. Frames are also defined as mental spaces that have become entrenched in long-term memory through repeated actions and experience (Fauconnier, 2007: 352). Hence, we can depict the mental association of the BLOOD-TAKING frame with the visual appearance of *Sessio* as part of a *conceptual blending* of two mental spaces (fig. 4).

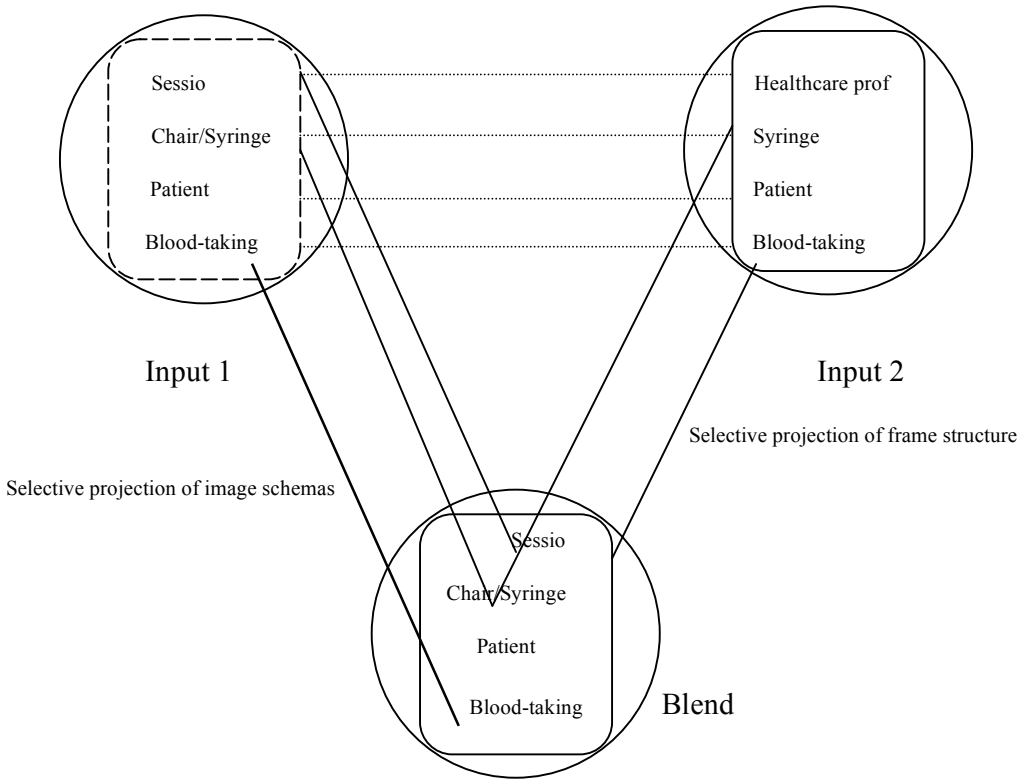


Fig. 4. Conceptual blending of mixed emotional content and frame structure

This so-called *network model* is trying to reveal that framing *Sessio* is not simply a matter of mentally projecting all elements of the existing BLOOD-TAKING frame (input 2) onto perceived form (input 1). For instance, *Sessio* is obviously not mistaken for a healthcare professional, but is conceived mentally as fulfilling the *role, job and intentions* of such a person. What the framing operation consists in is thus more specifically a 'selective projection' of a general *intentionality pattern* (indicated by the square). The job consists in getting a blood-sample by using a syringe. The intention is to diagnose symptoms of illness and ultimately to heal the patient. Whereas the role of the professional is usually paired with the ability to distance oneself emotionally from one's job, in the case of blood-taking patients do normally expect the agent to feel empathy for their situation.

The intentionality pattern available for robotic devices is not so richly structured (the dashed square) and even collides with the pre-existing BLOOD-TAKING frame at some points. Robots are not designed to be empathic about their job; they are mostly used for inanimate objects, not human patients; the intrusion of technology into the human body is not (yet) folk psychologically thought to cause healing, but often the opposite (cf.

the image of cyborgs, androids); and so forth. However, what the organic shape of *Sessio* invites patients to do is conceptually to leave this pattern un-activated and to blend only its physical elements with the intentionality pattern from input 2. The blend is a third mental space that makes composite structure available that did not exist in the separate inputs (Fauconnier & Turner, 2002: 42). On the basis of such conceptual integration new inferential and mental judgments can be made, which may of course elicit unexpected emotional responses: a cognitive conception of a considerate robot evoking surprise.

Blending at the Level of Embodied Interaction

So far I have uncovered some of the blending operations underlying appraisals of Appealingness and Novelty. What remains is to account for how the kinaesthetic and haptic feelings of the body equally contribute with organizing emotional structures to the blend. This is where embodied interaction and mixed emotions enter the picture.

The embodied interaction with *Sessio* differs from the standard act of tapping blood. Building the syringe instrument into the armrest of a chair means that *Sessio* is able to activate a bodily feeling that is normally absent. Remember that in the armrest the patient's arm is enclosed by vacuum. Being enclosed by something typically elicits one of two mutually exclusive bodily feelings. Either we feel claustrophobically confined, or we feel ourselves protected from external forces. I suggest that Sessio gives rise to the second bodily response, since we tend to feel safe and confident in those situations where larger parts of our skin is embraced in a soft and intimate way as in a hug. Yet, this bodily feeling changes into a sting of pain as soon as the needle is inserted. However, since the insertion and enclosure are enacted simultaneously, it is important that these two bodily feelings are seen as co-activated.

Johnson (1987) has convincingly demonstrated how *image schematic* structures underlie such kinaesthetic and haptic feelings of the body. To understand the co-activation of bodily feelings it might therefore be helpful to apply this concept. The surrounding of one's arm is experientially organized according to a schema of ENCLOSING as depicted in fig. 5.1. The open square along with the two arrows illustrates how *Sessio*'s armrest acts as a CONTAINER closing in on a body part (the circle). On the other hand, the insertion of a needle is evoking the sense of an ENTRY into a CONTAINER (fig. 5.2). Here, the needle is experienced to follow a penetrating path that creates its own opening into the body. In this instance the body is itself profiled as being the CONTAINER

(the square) of a contained object (a portion of blood; the small circle). Since all this is happening as part of the same haptic experience, it is reasonable to assume that the two image schematic structures are integrated at this embodied level (fig. 5.3).

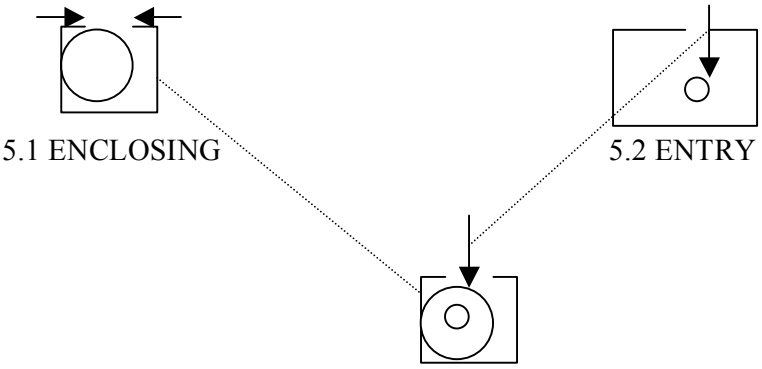


Fig. 5.3.

This image schematic analysis of embodied interaction offers a detailed structural description of the eliciting conditions of mixed emotions. Provided, of course, that we believe with Prinz (2004) that perceptual states arising from the detection of changes in bodily feelings are able to elicit emotions. Yet I claim that a proper understanding of *Sessio*'s emotional profile hinges upon this assumption of embodied appraisals.

In the actual use, it is not that one emotion is experienced as being superseeded by another emotion. Rather, one emotion is encapsulated by the other as the structural description clearly indicates. Having a needle inserted into our body is not experienced as pleasant, but since a pleasant embrace encapsulates this emotion it is at the same time downplayed or even counterbalanced.

One of the central assumptions of Fauconnier & Turner is that through conceptual blending image schematic structures of bodily experience might be integrated into the organization of higher-order mental representations. If we accept this line of thought, then fig. 5 could be seen as exemplifying the organizational structure that is projected into the blend from input 1 (the blood-taking line in fig. 4). What I am saying is that in the actual use patients are led through their embodied interaction to blend a new emotional response with the standard intentionality pattern of BLOOD-TAKING. As this blend is achieved, the intertwining of emotional and cognitive structures allows for building a new content into a reorganized concept – the emotional robot. This insight is crucial for understanding the

role of embodied interaction and mixed emotions in the reorganization of user expectation.

4. Conclusion

From this study of a 'bloody robot', it can be concluded that blending theory contributes to appraisal theories with:

- Detailed knowledge of the cognitive labor underlying already identified appraisal types
- The opportunity to incorporate embodied interaction into a framework for product emotions
- Detailed structural descriptions of the eliciting of mixed emotions by the appraisal of novel embodied aspects of technology design
- A structural model of the role that embodied interaction and mixed emotion play in conceptually reshaping user expectations

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